Moumen Nermine
nermine.moumen@insat.ucar.tn | +216 27 040 414

Subject: M2 Internship Spring 2027 — LIG Sensors

Dear Prof. Bedoui,

I am currently in my final year of the engineering program in Industrial Chemistry at the Institut National des Sciences Appliquées et de Technologie (INSAT – Tunisia), specializing in Process and Environmental Engineering. I am writing to apply for the M2 internship on laser-induced graphene sensors for volatile organic compound detection at UTC (Roberval), and I am available for 5 to 6 months starting February/March 2027.

My training has given me hands-on experience with Raman spectroscopy and SEM, developed during a literature review project on superconducting materials, along with pulsed polarography practice from an electrochemical treatment project. I also have solid practice with FTIR and NMR characterization, and computational experience in electrochemical device modeling (PEM fuel cells in COMSOL, PEM electrolyzers in Ansys Fluent) from my coursework and an internship at SEGULA Technologies focused on Li-ion battery modeling.

Your project, combining laser-induced graphene fabrication with electrochemical functionalization and machine-learning-guided processing for VOC sensing, represents a compelling opportunity for me to apply my materials characterization background to an innovative sensor development challenge.

This laboratory particularly appeals to me for its combination of hands-on materials engineering and data-driven processing, using structured datasets to guide fabrication rather than trial and error, within a collaborative framework spanning UTC and Universidad Javeriana Cali.

I would welcome the opportunity to discuss my application further and remain available for an interview at your convenience.

Thank you for considering my application.

Sincerely,
Nermine Moumen